

Appendix: Sample Lab Report

Acceptable SAMPLE LAB

Lab II, Problem 4: Projectile Motion and Mass

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Introduction

A group of medieval warfare enthusiasts is planning a reenactment and intends to build a trebuchet (Figure 1). If the reenactment is to be safe and realistic, the motion of the projectiles it launches must be well understood. Since the projectiles are rocks, their mass will vary so it is important to know where any rock thrown by the trebuchet will land. As a first step in doing so, this experiment sought to understand the acceleration of a rock after it is released by the trebuchet. The gravitational force on each projectile is its weight (W) which is proportional to its mass (m) and its direction is straight down, toward the center of the Earth.

$$W = mg \quad (1)$$

Where g is the proportionality constant and is approximately equal to 9.8 m/s^2 .

Furthermore, the acceleration (a) of an object is determined by the sum of the forces on it, Newton's 2nd Law.

$$\sum \vec{F} = m\vec{a} \quad (2)$$

If weight were the only force on the rock as it moves through the air,

$$mg = ma \quad (3)$$

and the acceleration would be independent of the rock's mass directed downward. However, adding air resistance as a force on the rock would change this conclusion unless the air resistance were also proportional to the rock's mass.

To test if the acceleration of a rock after thrown by a trebuchet is independent to the rock's mass in this experiment, the rocks were modeled using balls and the trebuchet, using an experimenter's arm. The procedure, data, and analysis of the experiment are given below. In this experiment the hypothesis that the acceleration is mass-independent was confirmed.



Figure 1: Picture of 3 trebuchets. Luc Viatour / <https://Lucnix.be>

Prediction

Assuming that air resistance is negligible when throwing a ball a short distance, Newton's 2nd law predicts that the acceleration of the ball would be straight down and independent of the ball's mass as shown in the introduction above. If air resistance is proportional to the mass of the ball, then the acceleration would still be independent of the mass of the ball, but it would not be straight down.

Procedure

Spherical balls, all approximately the same size to make the effects of air resistance for each ball the same, but of varying masses were used to model the rocks. The action of the trebuchet was modeled by one experimenter throwing a ball to another experimenter approximately 1.5 meters away. The resulting motion was recorded with a video camera using Vernier video analysis software to generate a horizontal position, vertical position, and time for each video frame. A meter stick was placed approximately 5 cm behind the projectiles' plane of motion for calibration. The position of each projectile as a function of time was analyzed to determine the velocity of the ball which was graphed as a function of time. The acceleration of each ball was determined as the slope of the velocity versus time graph.

The motion of seven different balls was analyzed in this fashion. The results are in Table 1 for the vertical acceleration. The accelerations in the horizontal acceleration were zero in all cases.

M in grams	a_y in m/s^2 (low)	a_y in m/s^2 (high)	a_y in m/s^2 (best fit)
48.8	9.7	10.7	10.0
51.4	9.3	11.1	9.5
57.3	9.0	10.6	10.0
75.0	9.0	10.0	9.7
141.2	9.1	10.5	9.8
148.6	9.3	10.8	9.9
165.5	9.4	10.5	10.0

Table 1: The vertical accelerations as measured by analyzing the vertical velocity vs time data from the video. The uncertainty in all the masses is 0.3 grams.

Analysis

The accelerations in the vertical (y) direction as measured by slope of the y velocity vs time graph are given in Table 1 in the Data section. The accelerations in the horizontal (x) direction are not given because they are all 0. Errors were assigned to the data points shown in Figure 2 by finding the maximal and minimal values of slopes which yield valid fits to the velocity data. The average of the accelerations listed in Table 1, was then taken as the single parameter in a 0-degree polynomial fit to the data. The error was taken to be the standard deviation from determining this average. The fit is depicted in Figure 2.

As Figure 2 illustrates, the fit of the data to a single value of the acceleration falls within the uncertainty of all the data points. This means that the uncertainty is likely to have been overestimated.

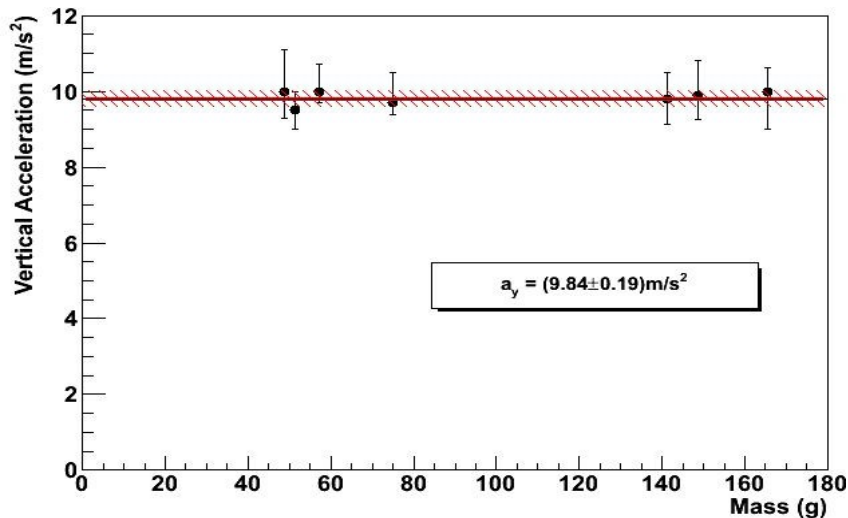


Figure 2: The measured vertical acceleration versus the ball mass and the fit of a constant to the data. The errors in the masses are smaller than the data markers.

Possible sources of systematic error include air resistance, distortion due to the camera's optics, and an error in position calibration due to the offset of the trajectories with respect to the meter stick. These, and any other systematics, are believed to be negligible because the average accelerations the vertical direction is consistent with the known gravitational acceleration at Minneapolis of 9.80535 m/s^2 based on measurements and the Earth Gravitational Model 2008 calculated by WolframAlpha.

Conclusion

The motion of projectiles launched by trebuchets was modeled by thrown balls. The hypothesis that their horizontal accelerations are mass-independent was confirmed in that all were measured to be 0. The hypothesis that the vertical accelerations are mass-independent was confirmed in that a single, constant acceleration of 9.84 m/s^2 that was within the uncertainty of all the measured data points. Although this conclusion is valid for balls thrown over a short distance, it may not be valid for rocks thrown by a trebuchet because air resistance will play a much larger role in that case. Further experimentation is called for where the balls are chosen to enhance air resistance. One such ball would be a ball that is a thin shell with holes in the shell such as a Wiffle ball.